SIMATS ENGINEERING

ASSESSMENT TOOL 1: AI Model Design Exercise

COURSE NAME: ARTIFICIAL INTELLIGENCE AND EXPERT SYSTEMS **COURSE CODE: MLA01**

COURSE OUTCOME COVERED
CO4: *Develop intelligent software agents using suitable agent architectures and learning techniques.*(BTL 3)
Assessment Tool Weightage: 40 %

Total Marks: 100 **Time: 90 Mins**
No. of Questions: 1

Annexure A

AI Model Design Exercise

Code of Conduct:
I G.Karthik Reddy-192525047 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: *G.Karthik Reddy*

Criterion	Weightage	Marks Obtained
1. Problem Analysis and Domain Understanding	10%	
2. Dataset Selection and Data Understanding	10%	
3. Data Preprocessing and Feature Engineering	10%	
4. AI Technique Selection and Justification	10%	
5. Model Design / Architecture	10%	
6. Algorithm / Pseudocode Development	5%	
7. Model Implementation and GitHub Repository	15%	
8. Experimental Design and Results	10%	
9. Performance Evaluation and Metrics	10%	
10. Result Analysis and Interpretation	5%	
11. Limitations and Future Enhancements	5%	
12. Documentation and Technical Presentation	10%	
Total	100%	

Signature of the Faculty Total Marks Awarded

ANSWER ANY ONE QUESTION

S.No.	Scenario / Problem Statement	AI Model / Technique to be Developed	Expected Development Outcome
1	**Smart Loan Approval:** A bank wants to automatically determine whether a customer is eligible for a loan using income, credit score, employment status, loan amount, and repayment history.	Decision Tree	Design and implement a decision-tree-based AI model and determine the most suitable attribute-selection measure such as Information Gain, Gain Ratio, or Gini Index.
2	**Student Performance Prediction:** A university wants to predict whether a student is likely to pass or fail a course using attendance, internal marks, assignment performance, study hours, and previous academic performance.	Inductive Learning	Construct a predictive model from training examples and derive generalized decision rules from the learned patterns.
3	**Disease Risk Prediction:** A healthcare system needs to classify patients into low-risk and high-risk categories based on clinical measurements.	Statistical Learning	Design a statistical learning model, train it using patient data, and evaluate its generalization ability on unseen data.
4	**Image Recognition System:** An organization wants to automatically classify images into categories such as vehicles, animals, and buildings.	Neural Network	Design and implement a neural-network classifier by specifying the network architecture, activation functions, loss function, optimizer, and training procedure.
5	**Handwritten Character Recognition:** A system must recognize handwritten characters from	Multilayer Neural Network	Develop a multilayer neural network and demonstrate how backpropagation updates weights to minimize classification error.

	scanned images.		
6	**Non-Linear Classification:** A dataset contains two classes that cannot be separated using a straight-line decision boundary.	SVM with Kernel Method	Design an SVM-based classifier using an appropriate kernel function and justify the selected kernel based on the characteristics of the dataset.
7	**Intelligent Recommendation Agent:** An online platform wants an agent that learns which recommendations to provide based on user responses.	Reinforcement Learning	Design an RL model by defining the states, actions, rewards, policy, and learning strategy, and demonstrate how the agent learns from user feedback.
8	**Autonomous Game Agent:** An AI agent must learn to play a game where successful actions receive positive rewards and unsuccessful actions receive penalties.	Reinforcement Learning	Develop an RL-based game agent and demonstrate how its policy and performance improve through repeated interaction with the environment.
9	**Explainable Diagnosis System:** A medical AI system predicts a disease but must also provide an explanation for its prediction.	Explanation-Based Learning	Design an explanation-based learning approach using domain knowledge and training examples to generate an understandable justification for the prediction.
10	**AI Model Selection Challenge:** A company provides a dataset for predicting customer churn.	Comparative AI Model Design	Design and compare at least two learning approaches from decision trees, statistical learning, neural networks, or kernel methods. Select the best model based on accuracy, interpretability, computational requirements, and generalization ability.

Structure of the Report:

S.No.	Report Component	What the Student Should Include
1	**Problem Statement**	Clearly describe the real-world problem, input data, expected output, and AI task to be solved.
2	**Objectives**	State the specific goals of developing the AI model and expected outcomes.
3	**Dataset Description**	Provide dataset source, number of instances, features, target variable, data types, and important attributes.
4	**Data Preprocessing**	Explain data cleaning, missing-value handling, encoding, normalization/scaling, feature selection, and train-test splitting.
5	**Selected AI Technique and Justification**	Identify the selected AI technique and justify its suitability for the problem.
6	**Model Design / Architecture**	Present the model structure, components, input/output, parameters, hyperparameters, and architecture/flow diagram.
7	**Algorithm / Pseudocode**	Provide the algorithmic steps or pseudocode showing model training and prediction/decision-making.
8	**Implementation**	Provide programming language, libraries/tools, important code sections, and implementation details.
9	**Experimental Results**	Present results using tables, graphs, sample predictions, learning curves, confusion matrices, or other relevant outputs.
10	**Performance Evaluation**	Evaluate the model using appropriate performance metrics such as accuracy, precision, recall, F1-score, MSE, RMSE, or cumulative reward.
11	**Result Analysis**	Interpret results, identify important patterns, compare actual and predicted outcomes, and discuss strengths and weaknesses.
12	**Limitations**	Discuss limitations related to dataset, model complexity, computational resources, accuracy, generalization, or deployment.
13	**Future Enhancements**	Suggest improvements such as additional data, feature engineering, hyperparameter tuning, alternative algorithms, or deployment.
14	**Conclusion**	Summarize the problem, methodology, major findings, model performance, and overall effectiveness.
15	**References**	List datasets, research papers, books, websites, documentation, libraries,

		and other sources used.
16	**GitHub Repository Link**	Provide the public GitHub repository URL containing the complete source code, dataset information/source, implementation, results, README, and supporting files.

Evaluation Rubrics:

Criterion	Weightage (%)	Excellent	Good	Average	Poor
1. Problem Analysis and Understanding	**10%**	9–10% – Clearly analyzes the real-world problem, objectives, inputs, outputs, constraints, and expected AI outcome.	7–8% – Understands the problem and objectives with minor omissions.	5–6% – Shows partial understanding with some important elements missing.	0–4% – Problem is poorly understood or incorrectly defined.
2. Dataset Selection and Understanding	**10%**	9–10% – Selects a highly relevant dataset and clearly explains features, target, source, size, and characteristics.	7–8% – Appropriate dataset with adequate description and minor omissions.	5–6% – Dataset is usable but description or relevance is limited.	0–4% – Dataset is inappropriate, poorly described, or missing.
3. Data Preprocessing and Feature Engineering	**10%**	9–10% – Performs appropriate cleaning, transformation, encoding, scaling, feature selection, and data splitting with clear justification.	7–8% – Performs most required preprocessing correctly with minor issues.	5–6% – Basic preprocessing is performed but important steps are missing.	0–4% – Preprocessing is inadequate, incorrect, or absent.
4. AI Technique Selection and Justification	**10%**	9–10% – Selects the most appropriate AI technique and provides strong technical justification based on the problem and dataset.	7–8% – Appropriate technique selected with reasonable justification.	5–6% – Technique is acceptable but justification is limited.	0–4% – Inappropriate technique or no meaningful justification.

5. Model Design / Architecture	**10%**	9–10% – Develops a clear, technically sound model architecture with appropriate components, parameters, and workflow.	7–8% – Model is well designed with minor omissions.	5–6% – Basic model design is presented but lacks technical depth.	0–4% – Model design is unclear, incomplete, or inappropriate.
6. Algorithm / Pseudocode	**5%**	5% – Algorithm/pseudocode is complete, logically correct, structured, and clearly represents the model development process.	4% – Mostly correct with minor omissions.	2–3% – Basic algorithm provided but contains gaps.	0–1% – Algorithm is missing or substantially incorrect.
7. Implementation and GitHub Repository	**15%**	13–15% – Fully functional implementation with clean, organized, reproducible code and a complete GitHub repository with README and supporting files.	10–12% – Functional implementation and well-organized repository with minor issues.	7–9% – Partially functional implementation or incomplete repository documentation.	0–6% – Implementation is incomplete/non-functional or GitHub repository is missing.
8. Experimental Design and Results	**10%**	9–10% – Conducts meaningful experiments with appropriate test cases and clearly presents results using tables, graphs, or visualizations.	7–8% – Appropriate experiments with clearly presented results.	5–6% – Limited experiments or basic result presentation.	0–4% – Insufficient, incorrect, or missing experimental results.
9. Performance Evaluation	**10%**	9–10% – Uses appropriate evaluation metrics and provides comprehensive quantitative analysis of model performance.	7–8% – Uses suitable metrics and provides adequate evaluation.	5–6% – Basic evaluation with limited metrics or analysis.	0–4% – Inappropriate metrics or performance evaluation is missing.

10. Result Analysis and Interpretation	**5%**	5% – Provides insightful analysis of results, identifies patterns, errors, strengths, weaknesses, and explains model behavior.	4% – Provides good interpretation with minor gaps.	2–3% – Basic interpretation with limited analysis.	0–1% – Results are not meaningfully analyzed.
11. Limitations and Future Enhancements	**5%**	5% – Clearly identifies realistic limitations and proposes technically relevant and feasible improvements.	4% – Identifies relevant limitations and reasonable improvements.	2–3% – Provides basic limitations and future suggestions.	0–1% – Missing, unrealistic, or poorly explained.
12. Documentation and Technical Presentation	**10%**	9–10% – Report is comprehensive, well organized, technically accurate, professionally presented, and properly referenced.	7–8% – Well documented with minor presentation or referencing issues.	5–6% – Adequate documentation with several gaps.	0–4% – Poorly documented, unclear, or incomplete report.
Total	100%				

Q2 – Student Performance Prediction

AI Model Design Exercise – Student Performance Prediction

1. Problem Statement

A university wants to predict whether a student is likely to **Pass or Fail** a course.
The prediction is based on:

- Attendance
- Internal marks
- Assignment performance
- Study hours
- Previous academic performance

The AI model learns patterns from previous student records and predicts the result of a new student. The original exercise describes this scenario as **Inductive Learning**, where a predictive model is constructed from training examples and generalized decision rules are derived from learned patterns.

2. Objectives

1. To collect student academic-performance data.
2. To preprocess the dataset.
3. To identify important factors affecting student performance.
4. To train an AI model using previous student records.
5. To predict whether a student will **Pass or Fail**.
6. To evaluate the prediction performance of the model.

3. Dataset Description

Feature	Description
Attendance	Percentage of classes attended
Internal Marks	Marks obtained in internal examinations
Assignment Score	Performance in assignments
Study Hours	Average hours spent studying per day
Previous Performance	Previous academic result
Result	Target: Pass / Fail

Input: Student academic information
Output: Pass or Fail
The exercise specifically requires the dataset section to include the source, number of instances, features, target variable, data types, and important attributes.

4. Data Preprocessing

The following preprocessing steps can be performed:

1. Remove duplicate records.
2. Handle missing values.
3. Convert categorical values into numerical values.
4. Normalize numerical features if required.

5. Separate input features and target variable.
6. Split the dataset into training and testing data.

Example:

Dataset
↓
Remove Missing Values
↓
Encode Categorical Data
↓
Feature Selection
↓
Train-Test Split
↓
AI Model

5. Selected AI Technique – Inductive Learning

Inductive Learning is selected because the system learns general patterns from previously observed student examples.

For example:

High Attendance
+ Good Internal Marks
+ Good Assignment Score
↓
PASS

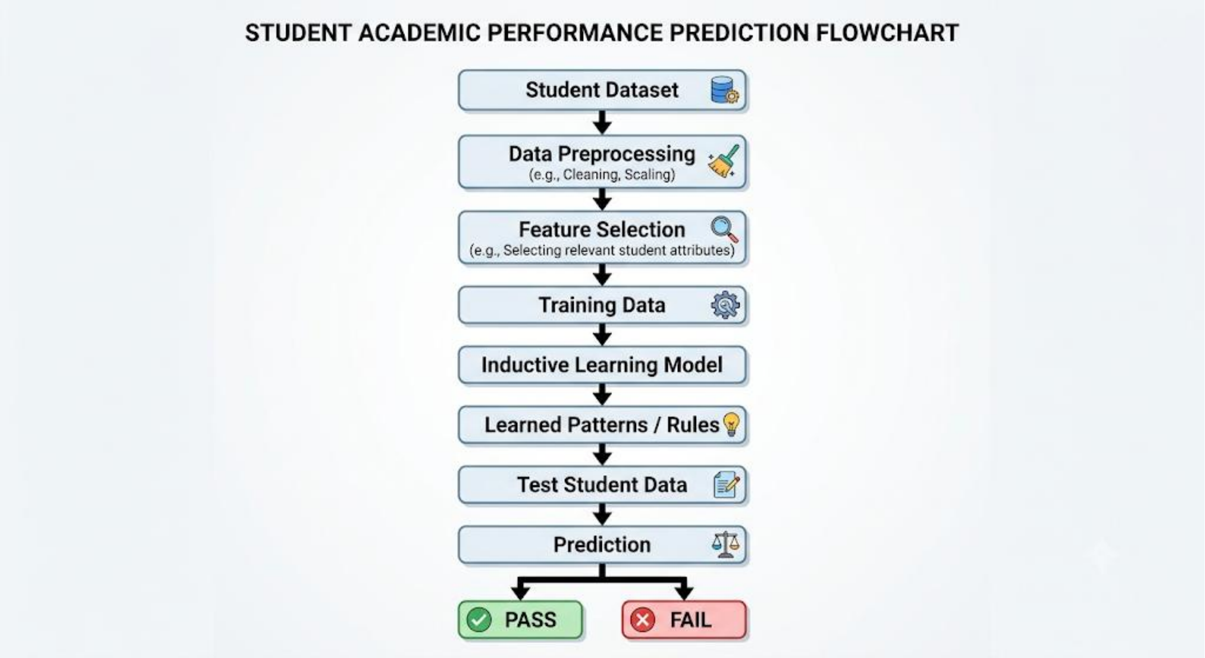
and

Low Attendance
+ Low Internal Marks
+ Poor Assignment Score
↓
FAIL

The exercise itself specifies Inductive Learning for this problem.

6. Model Architecture

A simple architecture can be represented as:



7. Algorithm / Pseudocode

START

1. Load student dataset.
2. Clean the dataset.
3. Select student performance features.
4. Separate features and target result.
5. Divide data into training and testing sets.
6. Train the inductive learning model.
7. Give test student data to the model.
8. Predict PASS or FAIL.
9. Calculate performance metrics.
10. Display the results.

STOP

The exercise requires algorithmic steps or pseudocode showing both model training and prediction.

8. Implementation

Programming Language: Python

Libraries:

- Pandas
- NumPy
- Scikit-learn
- Matplotlib

- Seaborn

A simple implementation can use a classification model to learn the relationship between the academic features and the student's final result.

9. Experimental Results

Example result table:

Student	Predicted Result
Student 1	Pass
Student 2	Pass
Student 3	Fail
Student 4	Pass
Student 5	Fail

You can present the results using:

- Prediction table
- Accuracy graph
- Confusion matrix
- Actual vs predicted results

The exercise specifically recommends tables, graphs, sample predictions and confusion matrices for experimental results.

10. Result Analysis

The model can identify patterns between academic factors and student outcomes.
For example:

- Students with better attendance may have better predicted outcomes.
- Higher internal marks can contribute to a Pass prediction.
- Assignment performance can help distinguish between Pass and Fail.
- Previous academic performance can provide useful information for prediction.

The actual importance of each feature should be determined from the trained model rather than assumed beforehand.

11. Limitations

1. Prediction depends on the quality of the dataset.
2. A small dataset may produce unreliable predictions.
3. Student performance can be affected by factors not included in the dataset.
4. The model may not generalize well to students from different institutions.
5. Incorrect or missing data can reduce prediction accuracy.

The exercise expects limitations concerning dataset quality, model complexity, accuracy, generalization and deployment where applicable.

12. Future Enhancements

- Use a larger student dataset.
- Add more relevant academic features.
- Apply feature engineering.
- Compare multiple AI algorithms.
- Perform hyperparameter tuning.

- Develop a web-based prediction system.
- Continuously update the model with new student data.

These improvements are consistent with the future-enhancement options specified in the exercise.

13. Conclusion

The Student Performance Prediction System uses inductive learning to learn patterns from previous student records and predict whether a student is likely to pass or fail. Attendance, internal marks, assignment performance, study hours and previous academic performance are used as input features. The trained model can support early identification of students who may require additional academic support.

14. References

You can include:

1. Scikit-learn documentation.
2. Python documentation.
3. Relevant machine-learning textbooks.
4. Dataset source used for implementation.
5. Research papers related to student performance prediction.

15. GitHub Repository

Your GitHub repository should contain:

```
Student-Performance-Prediction/
│
├── dataset/
│   └── student_data.csv
│
├── student_prediction.py
├── results/
│   ├── confusion_matrix.png
│   └── accuracy.png
│
├── README.md
└── requirements.txt
```